

1. Why Combine Demand and Supply?

- Neither demand nor supply **alone** determines the outcome in a market.
 - They must work **together** to set the price and quantity of goods bought and sold.
 - Prices act as **signals** — they adjust based on consumer demand and producer supply.
 - When demand and supply interact, the market moves toward a **balance** called equilibrium.
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2. Market Equilibrium

Market Equilibrium = the point where **quantity demanded = quantity supplied**, so there is no shortage or surplus.

Term	Definition
Equilibrium Price (P^*)	The price at which demand and supply are perfectly balanced
Equilibrium Quantity (Q^*)	The quantity bought and sold at the equilibrium price

On a graph: Equilibrium is where the **demand curve (D)** and **supply curve (S)** intersect.

How it works:

- If there is **excess supply** → prices fall → market moves toward equilibrium
- If there is **excess demand** → prices rise → market moves toward equilibrium
- Once equilibrium is reached → no pressure for price to change

Real-world example (Cloud Storage):

- Demand for cloud storage rises → shortage occurs → prices rise → providers expand → new equilibrium reached at higher price and quantity.
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3. How to Find Market Equilibrium (Mathematically) ★

The rule: Set $Q_d = Q_s$ and solve for P , then find Q .

Example 1 – Linear (Cloud Storage)

- Demand: $Q_d = 500 - 5P$
- Supply: $Q_s = 100 + 3P$

Step 1: Set $Q_d = Q_s$

$$\begin{aligned} 500 - 5P &= 100 + 3P \\ 500 - 100 &= 3P + 5P \\ 400 &= 8P \\ P^* &= 50 \end{aligned}$$

Step 2: Substitute P^* back into either equation

$$Q^* = 100 + 3(50) = 100 + 150 = 250$$

Answer: $P^* = \$50$, $Q^* = 250$ storage plans

Example 2 – Non-Linear (Smartphones)

- Demand: $Q_d = 300 - P^2$
- Supply: $Q_s = 100 + P^2$

Step 1: Set $Q_d = Q_s$

$$\begin{aligned} 300 - P^2 &= 100 + P^2 \\ 300 - 100 &= 2P^2 \\ 200 &= 2P^2 \\ P^2 &= 100 \\ P^* &= 10 \end{aligned}$$

Step 2: Find Q^*

$$Q^* = 100 + (10)^2 = 100 + 100 = 200$$

Answer: $P^* = \$10$, $Q^* = 200$ smartphones

Exam Tip: Always follow these steps:

1. Set $Q_d = Q_s$
 2. Solve for P^*
 3. Substitute P^* to find Q^*
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4. Market Disequilibrium

Disequilibrium = when quantity supplied $\neq$ quantity demanded $\rightarrow$ results in either a **surplus** or a **shortage**.

4a. Surplus (Excess Supply)

Surplus = $Q_s > Q_d \rightarrow$ happens when price is **above** equilibrium

Causes:

- Price is set too high $\rightarrow$ discourages buyers
- Overproduction by suppliers
- Decrease in consumer demand (change in taste, income, substitutes)

What happens next:

- Sellers lower prices to attract buyers
 - As price falls $\rightarrow$ demand rises, supply falls
 - Market returns to equilibrium
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4b. Shortage (Excess Demand)

Shortage = $Q_d > Q_s \rightarrow$ happens when price is **below** equilibrium

Causes:

- Price is set too low $\rightarrow$ attracts too many buyers
- Limited production / supply chain disruptions
- Sudden surge in demand (trends, new product launches)

What happens next:

- Sellers raise prices due to scarcity
 - As price rises → demand falls, supply rises
 - Market returns to equilibrium
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Quick Comparison: Surplus vs. Shortage

	Surplus	Shortage
Condition	$Q_s > Q_d$	$Q_d > Q_s$
Price is...	Above equilibrium	Below equilibrium
Market response	Price falls	Price rises
End result	Return to equilibrium	Return to equilibrium

Key point: In free markets, surpluses and shortages are **temporary** — prices naturally adjust to restore equilibrium.

Practice Example (Wireless Routers)

- Demand: $Q_d = 400 - 4P$
- Supply: $Q_s = 50 + 3P$
- Price set at **\$70**

Step 1: Find Q_d and Q_s at $P = 70$

$$\begin{aligned}Q_d &= 400 - 4(70) = 400 - 280 = 120 \\Q_s &= 50 + 3(70) = 50 + 210 = 260\end{aligned}$$

Step 2: Compare:

- $Q_s (260) > Q_d (120) \rightarrow$ **Surplus**
 - Size of surplus = $260 - 120 =$ **140 units**
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5. Changes in Equilibrium

When demand or supply changes, the old equilibrium is disturbed and a **new equilibrium** is established.

5a. Change in Demand Only

Scenario	Curve Shift	Effect on Price	Effect on Quantity
Demand increases	Shifts right	P ↑	Q ↑
Demand decreases	Shifts left	P ↓	Q ↓

Example – Demand Increases (Cloud Storage):

- Remote work trend → more people need cloud storage → demand shifts right → shortage at old price → price rises → suppliers expand → new equilibrium at **higher P and higher Q**

Example – Demand Decreases (Smartphones):

- Competitor releases better model → fewer people buy old model → demand shifts left → surplus at old price → company lowers price → new equilibrium at **lower P and lower Q**
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5b. Change in Supply Only

Scenario	Curve Shift	Effect on Price	Effect on Quantity
Supply increases	Shifts right	P ↓	Q ↑
Supply decreases	Shifts left	P ↑	Q ↓

Causes of Supply Increase:

- Decrease in input prices, technological improvement, more sellers, expected future price drop

Example – Supply Increases (Cloud Computing):

- New tech reduces data center costs → more companies enter market → supply shifts right → surplus at old price → providers reduce prices → new equilibrium at **lower P and higher Q**

Example – Supply Decreases (Laptops):

- Global semiconductor shortage → fewer laptops produced → supply shifts left → shortage at old price → prices rise → new equilibrium at **higher P and lower Q**
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6. Complex Changes – Both Demand AND Supply Shift ★ ★

When both change simultaneously, one outcome is **certain** and one is **indeterminate** (depends on magnitude of shift).

Case 1: Demand ↑ and Supply ↑

- **Quantity:** Always **increases** ✓
 - **Price:** Indeterminate (depends on which shifts more)
 - Demand increases more → P rises
 - Supply increases more → P falls
 - Equal shifts → P unchanged
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Case 2: Demand ↑ and Supply ↓

- **Price:** Always **increases** ✓ (both forces push price up)
 - **Quantity:** Indeterminate
 - Demand increases more → Q rises
 - Supply decreases more → Q falls
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Case 3: Demand ↓ and Supply ↑

- **Price:** Always **decreases** ✓ (both forces push price down)
 - **Quantity:** Indeterminate
 - Supply increases more → Q rises
 - Demand decreases more → Q falls
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Case 4: Demand ↓ and Supply ↓

- **Quantity:** Always **decreases** ✓
 - **Price:** Indeterminate
 - Demand decreases more → P falls
 - Supply decreases more → P rises
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Master Summary Table – All Changes in Equilibrium

Change	Price (P*)	Quantity (Q*)
Supply increases ↑	P ↓	Q ↑
Supply decreases ↓	P ↑	Q ↓
Demand increases ↑	P ↑	Q ↑
Demand decreases ↓	P ↓	Q ↓
Demand ↑, Supply ↑	Indeterminate	Q ↑
Demand ↑, Supply ↓	P ↑	Indeterminate
Demand ↓, Supply ↑	P ↓	Indeterminate
Demand ↓, Supply ↓	Indeterminate	Q ↓

Memory trick for complex cases:

- When both change in the **same direction** → Quantity is certain, Price is uncertain
 - When both change in **opposite directions** → Price is certain, Quantity is uncertain
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7. How Prices Allocate Resources

Prices act as **signals** that guide economic decisions and help allocate scarce resources efficiently.

Price Level Signal to Producers Signal to Consumers

High prices Produce more Buy less

Low prices Produce less Buy more

How equilibrium price guides the economy:

- Ensures producers make goods that are profitable (and therefore wanted)
- Encourages consumers to buy based on their needs and budget
- Directs investment to high-demand sectors

Example (Cybersecurity Software):

- Cyber threats increase → demand for security software rises → prices rise → companies (Norton, McAfee) invest more in R&D and hire more experts → supply expands → prices stabilize → everyone gets access to affordable security software.

8. Likely Exam Topics for Lecture 3

1. **Define** market equilibrium, equilibrium price, equilibrium quantity
 2. **Distinguish** between equilibrium and disequilibrium
 3. **Define** surplus and shortage — causes and how markets self-correct
 4. **Calculate** equilibrium price and quantity from given Qd and Qs equations (linear and non-linear)
 5. **Determine** whether a given price creates a shortage, surplus, or equilibrium — and calculate its size
 6. **Explain** what happens to P^* and Q^* when demand or supply shifts
 7. **Analyze** complex changes (both shift simultaneously) — identify what is certain and what is indeterminate
 8. **Explain** how prices allocate resources in a market economy
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9. 💡 Final Exam Tips

- For maths questions: **always set $Q_d = Q_s$** — never skip this step.
 - "Indeterminate" means the answer depends on the size of the shift — say this explicitly in essays.
 - For complex changes: first identify if it's "same direction" (quantity certain) or "opposite direction" (price certain).
 - **Surplus = price too HIGH. Shortage = price too LOW.** Remember this distinction.
 - In free markets, shortages and surpluses are **temporary** — prices self-correct.
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Good luck! 🍀